

ATHARVA SAMADHAN KHAIRNAR

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Education

Vishwakarma Institute of Technology (VIT), SPPU

Pune, India

B.Tech — Computer Engineering

2020 – 2024

- CGPA: 7.88/10.00, First Class

G.D. Sawant Junior College, MSBHSE

Nashik, India

Higher Secondary Certificate (HSC)

2018 – 2020

- Percentage: 68%, First Class

Army Public School, CBSE

Nashik, India

Secondary School Certificate (SSC)

2017

- Percentage: 92%, Distinction

Experience

[EDIT] Job Title

[EDIT] Dates

[EDIT] Company Name

[EDIT] Location

- [EDIT] Achieved X by doing Y, resulting in Z.
- [EDIT] Developed and implemented A, leading to B improvement.

Projects

Intelligent Document Question-Answering System (RAG & LLMs) | *Python, Streamlit, FAISS, Llama3*

- Developed an advanced **Retrieval Augmented Generation (RAG)** system for intelligent document querying, leveraging **Llama3 (via Ollama)** and **FAISS** vector search over PDF documents.
- Engineered a full pipeline including PDF chunking, **semantic search**, and embedding generation using **Sentence Transformers** for enhanced contextual understanding.
- Designed and implemented an interactive **Streamlit web application** for real-time document querying, providing an intuitive user experience.

Surface Crack Detection using CNN | *Python, TensorFlow, Keras, OpenCV*

[EDIT]2023

- Built a robust **Convolutional Neural Network (CNN)**-based image classifier for accurate detection of cracks on industrial surfaces.
- Implemented comprehensive image preprocessing techniques and conducted binary classification, achieving high detection accuracy.
- Trained and evaluated the model using **TensorFlow/Keras**, with detailed performance visualization and analysis.

Text Sentiment Analysis using RNN/LSTM | *Python, TensorFlow, Keras, NLTK*

[EDIT]2023

- Developed an **LSTM (Long Short-Term Memory)**-based sentiment classifier for categorizing text into positive, negative, or neutral sentiments.
- Utilized **Word2Vec/GloVe embeddings** for rich semantic representation and applied dropout regularization to prevent overfitting.
- Trained the model with Adam optimizer and cross-entropy loss, demonstrating proficiency in advanced NLP and sequence modeling.

Real-Time Threat Intelligence & URL Classification Engine | *Python, Scikit-learn, Pandas, NumPy* [EDIT]

- Engineered a real-time URL threat classifier using a **6-model soft-voting ensemble** (Random Forest, Logistic Regression, Gradient Boosting) to detect malicious/phishing URLs with confidence scoring.

- Developed a dynamic feature-extraction pipeline, incorporating URL entropy, suspicious keywords, risky TLDs, and structural patterns for robust threat analysis.

End-to-End Machine Learning Pipeline Implementations | *Python, Scikit-learn, Pandas, NumPy*[EDIT]2022

- Successfully implemented full ML pipelines from data preprocessing and feature engineering to model training, evaluation, and visualization across diverse projects.
- Key Projects: Titanic Survival Predictor, Breast Cancer Detection, House Price Prediction, Ad Click Prediction, Customer Segmentation, Loan Default Prediction, Wine & Iris Classification.

Data Shield — Automated Backup & File Monitoring System | *Python, OS Module, hashlib, shutil*[EDIT]2022

- Built a Python-based backup and monitoring tool utilizing **MD5 checksum comparison** to efficiently detect new or modified files.
- Automated scheduled backups with a command-line utility, providing real-time reporting and archiving capabilities.

Platform Surveillance System | *Python, psutil, OS Module* [EDIT]2022

- Developed a system monitoring tool to track CPU, RAM, disk, and network activity in real time using the **psutil** library.
- Automated periodic logging with a CLI utility, facilitating comprehensive system analysis and reporting.

Technical Skills

Languages: Python, C, C++, Java, JavaScript

AI/ML/DL Libraries: Scikit-learn, TensorFlow, Keras, OpenCV, Hugging Face Transformers, Llama3, FAISS, Ollama, Sentence Transformers, NLTK

Data Science Automation: NumPy, Pandas, Matplotlib, Seaborn, SQL, psutil, hashlib, shutil, smtplib, schedule, PyPDF2

Databases: MySQL, MongoDB

Tools Platforms: Git, GitHub, Jupyter Notebook, VS Code, Google Colab, Streamlit